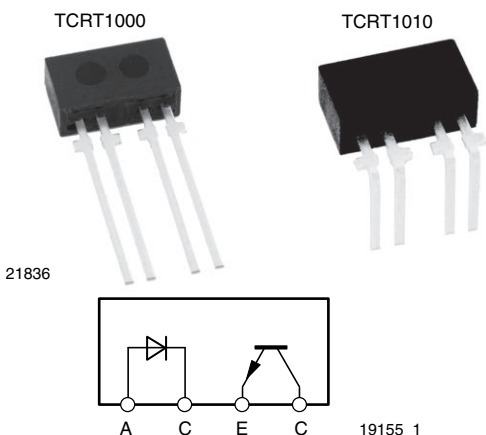


Reflective Optical Sensor With Transistor Output



FEATURES

- Package type: leaded
- Detector type: phototransistor
- Dimensions (L x W x H in mm): 7 x 4 x 2.5
- Peak operating distance: 1 mm
- Operating range within > 20 % relative collector current: 0.2 mm to 4 mm
- Typical output current under test: $I_C = 0.7$ mA
- Daylight blocking filter
- Emitter wavelength: 950 nm
- Lead (Pb)-free soldering released
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912


RoHS
COMPLIANT

DESCRIPTION

The TCRT1000 and TCRT1010 are reflective sensors which include an infrared emitter and phototransistor in a leaded package which blocks visible light.

APPLICATIONS

- Optoelectronic scanning and switching devices i.e., index sensing, coded disk scanning etc. (optoelectronic encoder assemblies for transmissive sensing).

PRODUCT SUMMARY

PART NUMBER	DISTANCE FOR MAXIMUM CTR _{REL} ⁽¹⁾ (mm)	DISTANCE RANGE FOR RELATIVE $I_{out} > 20$ % (mm)	TYPICAL OUTPUT CURRENT UNDER TEST ⁽²⁾ (mA)	DAYLIGHT BLOCKING FILTER INTEGRATED
TCRT1000	1	0.2 to 4	0.7	Yes
TCRT1010	1	0.2 to 4	0.7	Yes

Notes

(1) CTR: current transference ratio, I_{out}/I_{in}

(2) Conditions like in table basic characteristics/sensor

ORDERING INFORMATION

ORDERING CODE	PACKAGING	VOLUME ⁽¹⁾	REMARKS
TCRT1000	Bulk	MOQ: 1000 pcs, 1000 pcs/bulk	Straight leads
TCRT1010	Bulk	MOQ: 1000 pcs, 1000 pcs/bulk	Bent leads

Note

(1) MOQ: minimum order quantity

ABSOLUTE MAXIMUM RATINGS ($T_{amb} = 25$ °C, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
SENSOR				
Total power dissipation	$T_{amb} \leq 25$ °C	P_{tot}	270	mW
Ambient temperature range		T_{amb}	-40 to +85	°C
Storage temperature range		T_{stg}	-40 to +100	°C
Soldering temperature	2 mm distance to package, $t \leq 5$ s	T_{sd}	260	°C
INPUT (EMITTER)				
Reverse voltage		V_R	5	V
Forward current		I_F	100	mA
Forward surge current	$t_p \leq 100$ μ s	I_{FSM}	1.5	A
Power dissipation	$T_{amb} \leq 25$ °C	P_V	170	mW
Junction temperature		T_j	100	°C

ABSOLUTE MAXIMUM RATINGS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)				
PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
OUTPUT (DETECTOR)				
Collector emitter voltage		V_{CEO}	32	V
Emitter collector voltage		V_{ECO}	5	V
Collector current		I_C	50	mA
Power dissipation	$T_{amb} \leq 25\text{ }^{\circ}\text{C}$	P_V	100	mW
Junction temperature		T_j	100	$^{\circ}\text{C}$

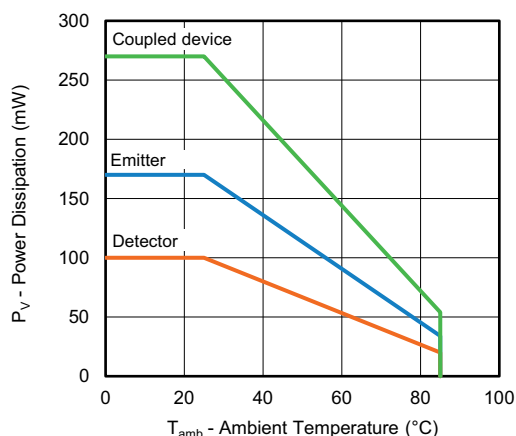
ABSOLUTE MAXIMUM RATINGS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)


Fig. 1 - Power Dissipation Limit vs. Ambient Temperature

BASIC CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)						
PARAMETER	TEST CONDITION	SYMBOL	MIN.	TYP.	MAX.	UNIT
SENSOR						
Collector current	$V_{CE} = 5\text{ V}$, $I_F = 20\text{ mA}$, $d = 1\text{ mm}$ (Fig. 2)	$I_C^{(1)}$	0.6	0.7	-	mA
Cross talk current	$V_{CE} = 5\text{ V}$, $I_F = 20\text{ mA}$	$I_{CX}^{(2)}$	-	-	1	μA
Collector emitter saturation voltage	$I_F = 20\text{ mA}$, $I_C = 0.1\text{ mA}$, $d = 1\text{ mm}$ (Fig. 2)	$V_{CEsat}^{(1)}$	-	-	0.3	V
INPUT (EMITTER)						
Forward voltage	$I_F = 100\text{ mA}$	V_F	-	1.6	1.7	V
Peak wavelength	$I_F = 100\text{ mA}$	λ_P	950	-	-	nm
OUTPUT (DETECTOR)						
Collector emitter voltage	$I_C = 1\text{ mA}$	V_{CEO}	32	-	-	V
Emitter collector voltage	$I_E = 100\text{ }\mu\text{A}$	V_{ECO}	5	-	-	V
Collector dark current	$V_{CE} = 10\text{ V}$, $I_F = 0\text{ A}$, $E = 0\text{ lx}$	I_{CEO}	-	-	200	nA

Notes

- (1) Measured with the "Kodak neutral test card", white side with 90 % diffuse reflectance
(2) Measured without reflecting medium

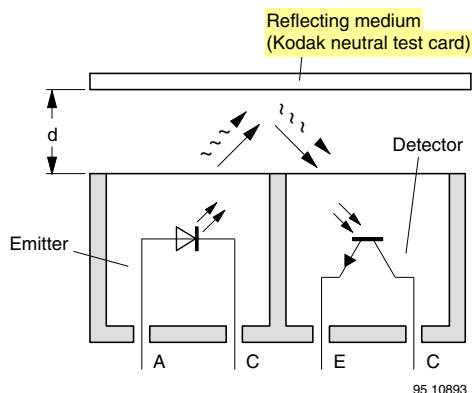


Fig. 2 - Test Condition

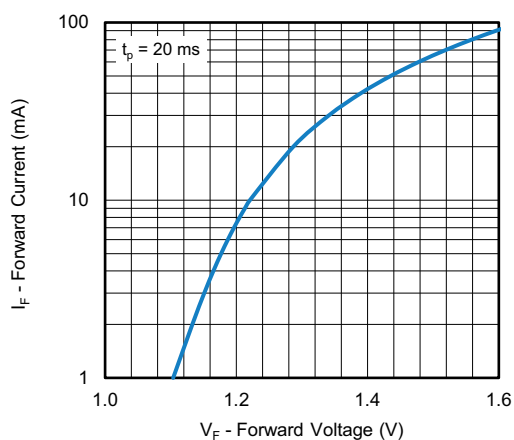
BASIC CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)


Fig. 3 - Forward Current vs. Forward Voltage

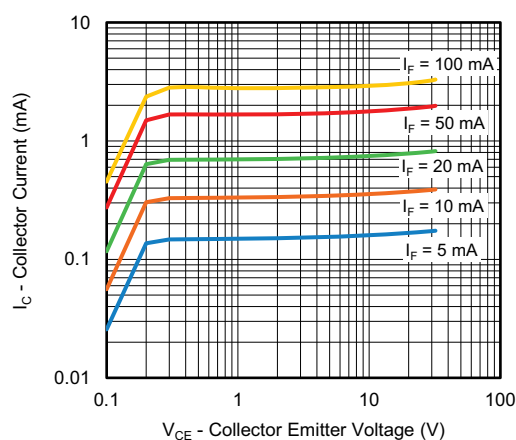


Fig. 5 - Collector Current vs. Collector Emitter Voltage

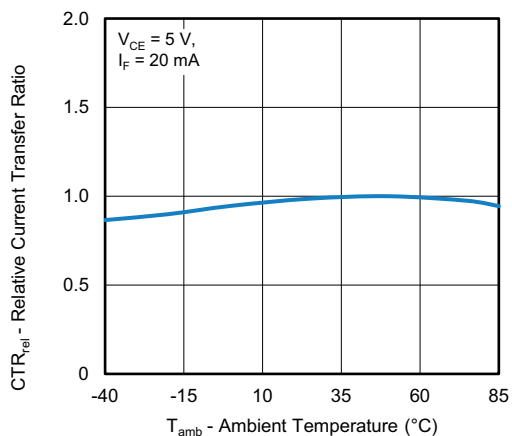


Fig. 4 - Relative Current Transfer Ratio vs. Ambient Temperature

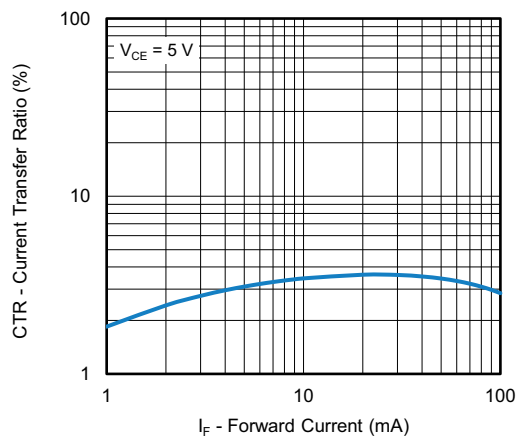


Fig. 6 - Current Transfer Ratio vs. Forward Current

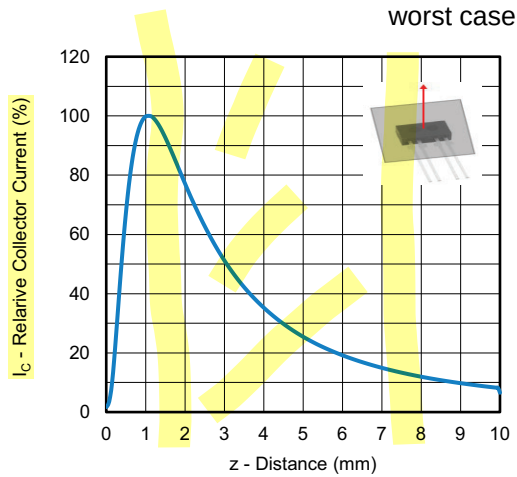


Fig. 7 - Collector Current vs. Distance

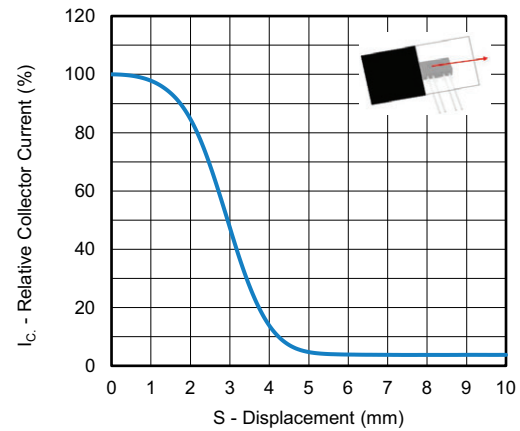
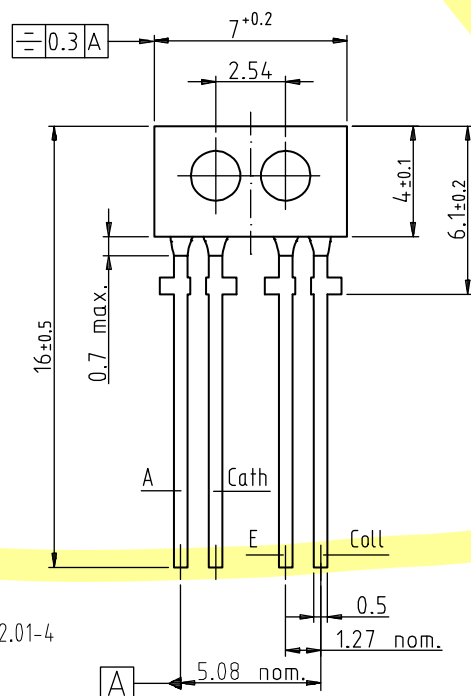
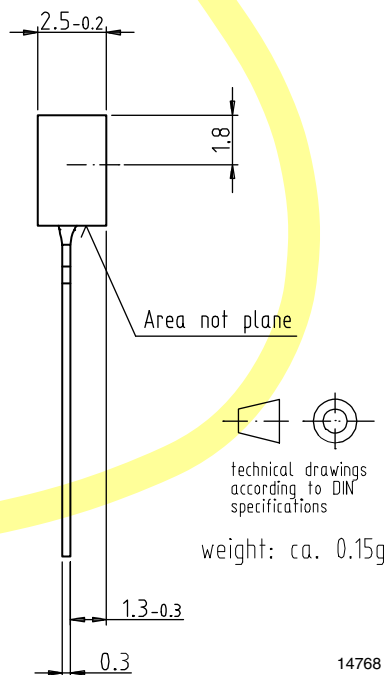


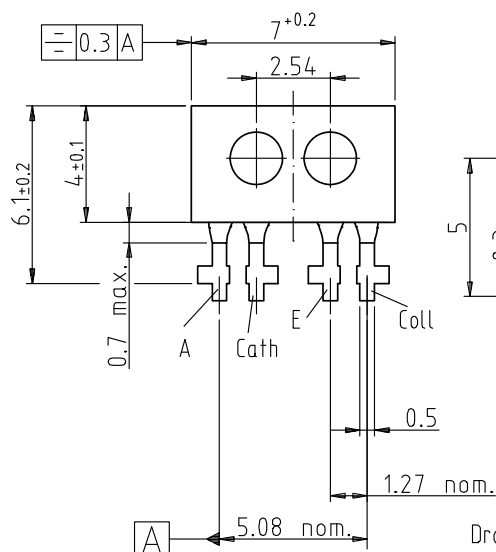
Fig. 8 - Relative Collector Current vs. Displacement

PACKAGE DIMENSIONS in millimeters

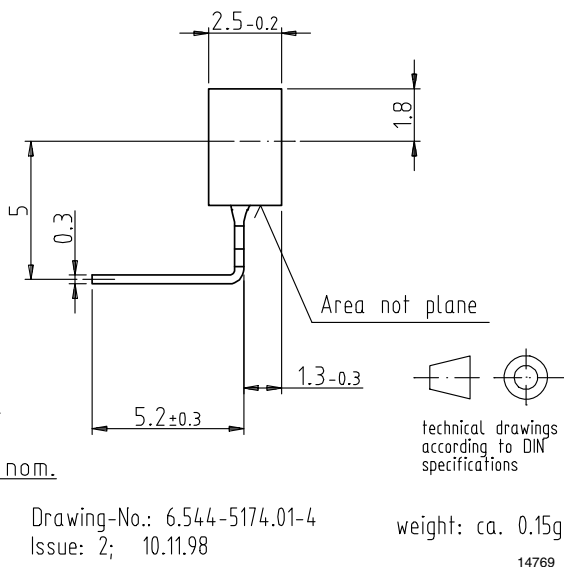
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Issue: 2; 10.11.98



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Issue: 2; 10.11.98



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